

Production/Operations Management is a discipline and profession that studies and practice the process of planning, designing and operating production systems and subsystems to achieve the goals of the organization.

1] It deals with process who can come under- production schedules quality assurance supervisors, materials managers, department supervisors, store manager and those who manage any personnel equipment or materials perform operations management tasks although they may not be primarily operation manager.

2] Management – it should be interpreted broadly to include designing the system and performing all activities necessary to operate the system, including directing personnel and acquiring materials and equipment.

3] Subsystems – need not be managing the production & organizations trial products, can be those providing products to other parts of production system.

E.g. cost accounting dept. provide reports and accounting information to other units of the company this does not go into final product, but cost accounting is a production subsystem, and its manager is also an operations manager though title may be cost accounting manager.

4] Definition does not say goal is to minimize costs. Many organizations have more than one goal- efficiency/productivity/cost reduction are important considerations; cost minimization is not a principal goal Primary goal of most businesses are to earn a steady and growing stream of profits and to maintain long term demand for their products as to stay in existence. Decisions made by operations manager influence revenue part of profit equation and cost, long term demand of products. Mercedes Benz has higher cost per automobile then Hyundai/Honda but that does not mean their production systems are superior to Mercedes Benz.

Not for profit organizations commonly have as their goal to deliver cost-effective levels and types of services.

Providing the lowest cost education or health service may not be desirable for the organizations or society.

Example what encompasses the operations management Price Waterhouse Coopers LLP:

Monitor existing work for the customers and assign new work to the staff under, as it is received. The unit under the manager has the human resources, training, equipment, and technology needed to provide customer high quality products in

a timely manner. Apart from these day-to-day operations need to give time for strategic issues evaluating new tools and technologies, designing new products restructuring the production process, forecasting sales and staffing needs throughout the country/area, and designing jobs so that the staff can progress to positions of leadership. Overlaying all these activities is constant attention to high quality in the products delivered to customers and the work done in the company.

What is the role of operation / production manager?

For practice: : Large manufacturing >10,000 crores

: Medium manufacturing 500 crores

: Small manufacturing 50 crore

Accounting firm

Lawyer firm

Chain of hospital / single hospital

Chain of Resorts / Single Resort

Food outlet

Dept. stores.

Modern Approach : modern operations management began in the industry revolution England where technologies advances were used in the way products were made , scale of operations and the organizational structure of personnel (Textiles) earlier example: building the wall of China – solving many operational problems, material acquisitions, transit, personnel training, scheduling, facility locations and the organization of work.

After Textiles-machine Tools-specialization of labour got into operations with scientific discoveries we had a structure of scientific inquiry of what later was

called scientific management –to design jobs better, train workers provide financial incentive to increase worker productivity. World War II through military operations-> operations research (OR),use of mathematical model in the industry → management science (MS).

These were new methods for operation management which involved personnel + production scheduling, vehicle routing and logistics facility location capacity planning facility design, queuing, inventory planning and statistical quality control. Development in computers helped in both OR/MS methods followed by flexible manufacturing systems-computer aided design to speedup product development and mainly better information flow production jobs and material can be tracked throughout the production process & information passed quickly across all the departments and to customers.

Then came Japanese production system which dominated many industries (US used in World War, taught Japanese later & Its influence in US diminished after the war, whereas Japanese adopted & improved upon what we called as Japanese Production system It is based on a)Quality comes first b)improve the product and process continuously and c) eliminate all forms of waste.

The basic principle and strategies for solving operations have existed for many years but now we have better tools and information to execute them. We now need environmentally compatible yet efficient systems-reduce harmful by products- recycle waste material + energy, reduce packaging use closed water system for cooling and waste discharge, scheduling work hours to reduce traffic and air pollution. Many companies do find using environmentally sound production methods – socially responsible economically beneficial.

Manufacturing systems-Production systems that produce goods are referred to as manufacturing systems and production of goods is manufacturing examples chemical, automobile, steel, computers airplanes, beverages, packed food, furniture. Services are intangible products that satisfy some need of a consumer including the enhancement of Goods. Production systems that produce services are referred to as service systems. Healthcare legal assistance financial services accounting services, educational instructions personal transportation-examples. They enhance consumers life in some way and serve as a useful function but cannot be touched or stored. Products can be a combination of goods and services:

Product	Tangible component	Service Component
Restaurant Meal	Prepared Food	Delivery to table clearing of dishes, ambience

Producer financed Airplane Financial services – Loan
Airplane

Turnkey Computer Computers, Software installation, training,
System/network maintenance

Home remodelling New Room Design Consultation

Manufacturing System

Batch/Intermittent			Continuous	
A]Unit	B]Jobs	C]Bulk	D]Mass	E]Process
Single item	Few Items	Bulk item	Standard Numerous items	High volume, Automated
Very large equipment, made to specific design	Several equipment made to same design	Paints, Specialty chemicals	Assembly Line consumer appliances, filling, packing	Cement fertilizer petrochemicals refineries

A] From single plant :- Produce for a batch-> factors of performance are: batch time, production per batch, product yield, specific utilities consumption and operational reflexes towards quality, safety and environment, utilities and operational reflexes towards quality, safety, and environment.

B] For Multiple operations / teams produce a batch – all above factors are applied individually and an individual also is judged by operational reflexes, ability and effort made.

C] When a batch is shared between shifts performance at front end and back end then individual's operational reflexes, ability, effort are considered. Also, it is linked to production of output – units produced per unit time (day/month/year), yield of key product per unit input [key raw material], specific consumption of utilities.

D] Assembly line / shop floor- even with automation influence by workers level of skill and capacity to make effort individuals output may vary. Performance rating is the basis for SALARIES, BENEFITS, Incentives, promotion to regular workers also for contract workers.

Plant performance – units per person per shift units rejected per shift units of power consumed per unit product.

E]Continuous process highly automated, need operator /supervisor attention for manipulation of operating conditions where automation problem is seen, standby availability in case of emergency to take actions during shift monitor /control the production rates and quality of products guide product transfer raw material scheduling and round the clock coverage, performances- sustenance of production rates, product yield, specific raw material and utilities consumption, plant wastes, performance targets or benchmarks, operational reflexes towards quality, safety and environment.

Interface management – it means management of common boundaries among people, systems, equipment, and concept. It intends effective communication and coordination across domains phases and activities which are interdependent.

Suppliers' product, buyers, technical services, utilities, supplies, quality control, materials, human resources, maintenance, Statutory bodies, corporate management.

Benefits of interface management – establishes a deep understanding of job /project complexity, optimizes resources / designs/quality/compatibility/cost/risk and schedule.

-Improves planning and reducing waste, builds/maintain relationships, and communication channels, ensure reliability of interdependency within increased transparency, identifies re-engineering and value additions. Key factors: no gaps in communication, no overlaps in roles and responsibilities, clear work definition, clearly defined responsibility, no buck passing, working with updated work matrix, conducting interdepartmental reviews.

Manufacturing / operations strategy –

Principles and concept organizations strategy is the map, compass, and rudder for the organization. It tells members of the organization where it wants to go and how to get there, it provides a mechanism for steering the organization.

Each functional unit marketing finance operations etc. should have its own strategy which is designed to support the organization strategy and to enhance the ability of the organization to achieve its goals.

Earlier marketing strategy was main for most of the business and operations function was to have a minimal strategic role it was supposed to produce and deliver the products specified by the marketing function at minimum cost. Decisions about operation can enhance organization ability to compete for customers with respect to price quality responsiveness and reliability of delivery product, customization, and flexibility.

Unless operation strategy inconsistent with the organization strategy even good operational performance can harm the organization.

Example: if a company markets a product based on quick delivery and product customization but the operations function is focusing on product standardization and low cost then the product may not sell even if the operations are successful in keeping costs low.

Equipment and production processes can now be selected and economically justified as they can reduce product delivery time or increase the range of products that can be produced, so improving the firm's ability to attract and retain customer rather than they reduce the cost of making existing products. Evaluating alternatives with respect to strategic goals and not simply cost, must be a part of all operations decision making. The operations strategy should guide decision related to technological capabilities, facilities, processes, quality, and personnel decision related to production management. Marketing and operations strategies must be consistent and complimentary as market environments are constantly changing so mechanism should exist for reviewing and revisiting strategies on a regular basis.

Operations strategy should include explicit sub strategies or plans for those aspects of operations that play an important role, example- Barnes and Noble which competes in terms of large selection of books carried should have explicit sub strategies for facility, capacity, design, and inventory management.

The operations strategy needs to be proactive instead of reactive, the characteristic and competitive advantages of the operations function should influence which new products to make which markets to serve and along which dimension the company can compete more effectively.

Operation as a competitive weapon-

Following will show how operational strengths can be used effectively as competitive weapons.

1]Product /Process expertise – company can have special expertise in making certain products e.g.: Sharp Corporation in consumer electronics due to dominance in LIQUID CRYSTAL DISPLAYS, DETROIT's Hyatt Regency Hotel had Japanese speaking employees to assist Japanese Customers (Detroit- Automobile and steel - Japan hub for cars).

2] Quick delivery - one can have a production process and capacity to produce a product and satisfy customer's request quickly- one hour photo, same day dry cleaning, shoe repair.

3] Short product cycle – 1st company to bring a product into the market gets significant market share. It depends mainly on production system HP [HEWLETT PACKARD] in laser printers, 1st to enter and continued to bring new products quickly before its competitors.

4] Production flexibility - highly flexible and rapidly responsive operations- Celestica Canadian computer component company - equipments used are not bolted to the floor so reconfiguring production lines take much less time to make new and different products Dell computers used build to order production systems that gave delivery of customized PCs in 5 days of ordering.

5] Low-cost process - A COMPANY with efficient production system or access to low-cost resources can make standard products at lower cost than its competitors. Example: steel companies with mini mills that uses scrap steel than larger integrated steel producers who make primary steel from iron ore. Also, these mini mills make limited commodity products than wide range a larger steel producer will. Southwest Airlines in USA is a low-cost passenger airline.

6] Convenience and location - Amex (card company) used to have offices located throughout the world to replace lost or stolen travellers' cheque and provide other travel services than its competitors. It could charge higher fees for many of its services.

7] Product variety and Facility size - grocery stores, Super store, retailers compete by having larger stores that carry large inventory of products to have additional services like video rentals and banking services.

8] Quality – higher quality commands higher price, Toyota and McDonald though initially started as low price for order winning, worked hard to develop a reputation in high quality and then could demand higher price than competitors. McDonalds – successful at achieving consistency with respect to food quality quick service and cleanliness of facilities.

Investment strategy: provision for future growth - Long-term investment in medium return companies, short term investment in high return companies, accuracy in project cost estimates, invest only for assured markets, use only well-read feasibility studies .

Capacity Strategy: helps maximize markets share compared with demand and supply, even out abrupt changes in demand, runs risk of building large unsupported capacity, not building enough capacity mean opportunity lost, focus is on product life cycle.

Company will need to consider both, the capacity of the entire system and the capacity of individual facilities.

- add capacity before it is needed so can satisfy exceptional surge in demand
- let capacity lag behind demand maximizing capacity utilization

- match capacity to demand over time
- add capacity on a regular basis irrespective of short-term demand changes.

Also consider a] economies of scale v/s distribution costs- 1 million units per year of a product

A] one large facility of 1 million or b] 10 units of 1 lakh each spread over an area that is meant to the customer which cost will be lower dictate the capacity.

B] lumpiness of capacity – some cases one can add in large increments e.g.: to steel – 1 million tons per year addition makes sense than just 50000 tons per year new hot rolling mill need 1-million-ton capacity

C] product variety and customer service - some cases like grocery stores, toy stores, hospitals, hotels, large capacity can provide better customer service with lower costs. Barnes Noble → large selection of books within store café, they made large stores than typical bookstores. Hotels competing for large conventions – number of sleeping rooms and meeting ROOMS THOSE WITH INSUFFICIENT number get eliminated from consideration.

Quality Strategy: Total Quality Management- a) quality is customers focused and customers driven, b) there is an organizational statement about quality, c) everyone in the organisation is responsible for quality and customer service, d) viewed quality improvement is complementary to efficiency improvement and not a hindrance, e) employees are treated with respect, empowered to make suggestions and changes to improve product quality and efficiency. It is to state the role of product quality in the overall strategy and how it is link to marketing: a) standard and good practices in place, b) adopting a modern process, c) buying quality materials, d) sell quality products avoid quality violation.

Technology Strategy: A company's technology can be defined as an equipment people and procedures used to produce products. So, a company's technology capabilities and the ways in which it uses technologies are important Technological strategies.

It is important to know what questions to ask and understand how technology fits into the organization strategy –

- a) sustainable and cost effective,
- b) constant upgradation safe and serviced,
- c) regular interval audit,
- d) optimum automation,
- e) maximum process integration,
- f) maximum modular approach.

Basic questions - what a company should do itself and what it can outsource:
example- calculator Bowmar Instruments, originator of handheld calculators was a product designer and assembler of components. It did not make electronic components, which is the heart of a calculator. It relied on external companies. After an initial 3 years of success, it started losing and Texas Instruments entered as the market grew and they could offer calculators at low prices and include new advances in their products. Bowmar tried to integrate backward into component manufacturing, it did not have technical expertise and went bankrupt.

Example: Corning used its expertise in making glass for sunglasses and auto headlights in liquid crystal displays. Proprietary processes are powerful weapons – for example: HP expert in computers, computer networking, test, and measurement instruments, to go for Diagnostic equipments.

For example, Ford, Walt Disney – deepest and broadest bases of any service industry. Animated movies, theme park attractions are conceived, designed, theme parks have outstanding industrial engineering staff that are known to be world leader in queueing facility layout, people management.

- adopted similar advances to their hotel technology throughout the organization - not only primary production but also in sub systems, typically, in service organizations,

a) electronic data interchange - link customers with suppliers,

b) point of sale cash registers- retail companies,

c) barcodes, optical scanning technology to track material.

Customer focus: a) well aware of customer needs, b) feedback mechanism in place, c) marketing for customer satisfaction, d) regard customer interface, e) prompt post sales service.

Facility Location Strategy: Location directly affects the cost of producing products, sales that they generate and even whether they will have access to certain markets. How close the company should be: close to the customer where the cost of production will be low, some suitable location is not important: example – Toys – as the criteria for purchasing a toy does not include where it is made within a country. Service companies such as grocery stores, restaurants, fuel stations, theatres - location is especially important, for example – McDonalds are location leaders.

Companies using a follower strategy wait for a competition to open a facility and then they open nearby- it reduces evaluation and search cost and reduces risk of operating in undesirable area.

International location - globally oriented companies extending their focus in other countries to increase their market sales allow economies of scale more fully:

- a) land- freehold, free of religious tangles, development cost,
- b) proximity to labor, raw material, water, and power,
- c) market dynamics – movement, shifts,
- d) the surrounding serenity, climate, neighbouring structures,
- e) services – medical, restaurant, school/colleges, entertainment,
- f) Roads – network, conditions, traffic width, destinations,
- g) sewage system - public effluent,
- h) local labor – attitude, skill level, economic condition,
- i) connectivity – transport, bus/railway/airport, nearest city,
- j) cost of transportation,
- k) industrial cluster,
- l) duties, taxes.

Product Flexibility Strategy: variety and volume of product, a company can produce:

- a) project processes which have maximum flexibility, make unique products such as a building, a computer program, financial audit, a Space Shuttle.

They are limited in their ability to exploit efficiencies of repetitiveness and specialization and flow processes which are designed to produce a specific product or narrow range of products in large volume at low cost.

These are highly automated, use specialized equipment and job designs. They have limited flexibility in producing a wide range of products and they cannot respond to change in product quickly.

Auto assembly, electronics/computer/car wash - example of repetitive processes, paper manufacturing, water processing, packing operations,

- c) in the middle we have job shops, batch flow processes, as these produce a wide variety of products in a low to medium volume products – a group of cutting machines in a factory, x-ray unit in a hospital.

These have considerable flexibility, can quickly lead to a change in product.

Quick time delivery Strategy: a) satisfy customer specific needs, b) match the process and product specifications, c) optimize plant and process runtime, d) implement efficient logistics, e) track movement by using Global positioning system GPS. It is a production process and capacity that allow the company to produce a product and satisfy customer's request quickly - one hour photo, same day dry cleaning, shoe repair.

Short delivery process - typically observed in a batch process, here a variety of products are normally made, and many are made to order.

Time required to process each job or product at each workstation, work center or work cell varies from job to job, because of differences in set up and processing requirements and in customer order sizes. As products are made in batches and stages of production are disconnected, each stage must be scheduled separately but coordinated with the other stages.

The shortest delivery process is taken up first by the equipment or a setup and the sequencing is done as per the time required for each process. The one which has the shortest product cycle should be taken a first, to take the opportunity it offers.

Process strategy - is a company's documentation approach that helps to achieve goals and transform resources into goods and services. By focusing on the products that require short delivery, based on the market, a company can grab a market share, be a leader in that product or services.

A short delivery process means speed in operation and improvement accelerating changes in technology, competition, regulations, and globalization demand that decisions get made faster at all levels.

Example - Google's engineering culture - they run lots of experiments with new product of feature ideas and let the market decide which one deserves further investment. It will look like chaos from the outside, but they are not afraid to fail fast and learn to scale up quickly if an idea shows merit. They have 20% time projects- engineers are expected to spend 20% of the time on projects that are creating and testing innovative ideas. This is open technology, meaning that workers at every level feel they can have significant impact.

Concept of productivity

Productivity can be defined as the amount of output produced divided by the amount of input used. The greater the amount of output from a fixed quantity of inputs, the higher is the productivity. So also, smaller the quantity of inputs for a fixed quantity of outputs. For several inputs and output, it becomes difficult to combine them into common units of measure. Example - different capacity bottle produced per unit time, similar product but at different quality, car models, legal cases, for the product is unique to each customer.

Measurement of productivity – a) output/labor ratio,
units of output produced/person-hours labor used: example - production systems making commodity type products where raw materials are same for all products. The steel industry tons of steel per person-hour or its reciprocal: example - individual machines or departments performance overtime or individuals doing same task - number of bottle filled/person-hour. This does not measure contribution to achievement of the organization goal, example - counterproductive if this is followed: one operator plus one or two people to remove and pack the product the product which comes out of the machine: One Packer is equal to 8000 units per hour can be produced, $\text{output} = 8000/2 = 4000$. 2 packers = 10,500 units/hour by increasing machine speed, $\text{output} = 10500/3$ is 3500, but extra capacity of 8000 to 10500 per hour can eliminate additional production facilities or 2nd packer can make an order profitable in a short term or faster rate will allow quicker delivery.

b) output value/labor ratio: monetary value of output produced per person-hour, here it omits other production inputs such as equipments, raw materials. One can increase output value/labor ratio by using higher quality materials which

generate higher revenue or transfer component production to other companies to reduce its own labor input, but these may not increase profit always, c) value added/labour ratio- how much additional value is created by the company's workers not those of its suppliers. The problem can be here using monetary value as short term changes in prices, the productivity will change though there is no change in the production process. These are defective measures when output or value amount produced rather than the amount shipped, as production departments will produce more than is needed, may produce the wrong mix of products as it shows the department more productive. Another danger in using productivity measures at the division or company level is that they can be improved by shrinking operations, short term productivity looks good but long term profitability is doomed.

Improvement: it is necessary to have useful information about the cost and profitability of various products and processes: example - creating profit centers or cost centers within the operation function. It is important how to allocate the costs. Many costs incurred by the organization are not easy to assign to specific cost centre and product, all may benefit from R&D, corporate finance, personnel, accounting, information systems, engineering services but vary considerable from department to department. For accounting simplicity, those costs that cannot be tied directly to a sole product are often pooled and allocated to all relevant products and the centres based on some measure of usage as direct labor-hour, machine operator-hour or units of product process. But using allocated cost in decision making can be dangerous. High allocation of workload and indirect costs based on direct labor hours lead to misdirected operational effort, managers will start reducing direct labor hours but in reality it will not reduce overhead and indirect cost of the company, example - to reduce direct labor cost one will do understaff or not, train workers adequately, this will be disruptive and incur long term total costs and hurt revenue.

Activity based costing gives a direct link between indirect overhead cost and the products made.

Goal based measures of performance:

- (i) long term goals require measures of long term performance,
- (ii) measures should reflect revenue impacts as well as cost impacts where appropriate,
- (iii) measures should evaluate organization wide impact (decision in one department can affect another),
- (iv) different measures are needed for different organizational levels.

Following are performance measures:

- a) quality- poor quality reduces capacity, waste resources delay deliveries so each operation unit should be evaluated with respect to its quality of work,
- b) on time delivery - late delivery affect future costs and revenues, due dates are important, and work done by them that/before that,
- (c) employee turnover - higher number is higher training costs, low productivity, poor quality,
- (d) safety - number of accident/incident free man-hours,
- (e) technical capability - subjective evaluation of the firm's ability to introduce new products quantitative measures of the time needed to introduce new products can be used, (f) machine breakdowns - measures of machines availability promotes better maintenance and wise use. More comparative a product market is, more important it is for a company to distinguish itself from the competitors in some order, winning dimension, incremental improvement in quality as on time delivery can separate out a better company and increase its sales.

Lean manufacturing: Japanese organization were instrumental in its development. The entire way products were designed, work was organized, and responsibilities were assigned, was transformed and constant striving for

improvement with elimination of waste was instilled. It is not of more automation, leading to an inflexible nature, but simplify the process and make it more flexible, it is complete re-engineering of the production process that emphasizes continuous improvement, quality management, reduced setup times, improve maintenance procedures and cooperation with supplier.

Reduce set-up times –

a) observe and analyse how the setup is currently performed - can make use of videos to understand actual activities,

b) separate internal and external setup activities - write a setup plan that shows the external setup activities as performed while the process is operating,

c) convert internal to external setup activities - changing work methods adding work-aids or buying duplicate sets of tools or equipments can assist in this conversion,

example - addition to a reactor and heating to a temperature can be changed to, add preheated chemical - internal to external which can be done parallel and not in series,

- simplify and streamline activities:

A) each workstation has its own tools and store the tools conveniently, example - technician in a medical lab required to recalibrate weighing scales before each test, this required three standard weights, if each technicians have its own set of standard weights, does not have to wait between batches to use standard weight share,

B) standardize sizes and shapes of equipments/parts used in a set up. Reliable delivery of materials, customers and suppliers can work together to provide reliable delivery to the customer at lower cost to both.

a) Share production scheduling plans quickly, any changing scheduled production volume and product mix need to be notified immediately to the suppliers,

b) include suppliers in product design - leads to components supplied are easier produced by the supplier, more consistent in quality and lower cost,

c) help suppliers to improve the production methods,

d) have specifically choose facilities - delivery time do matter, short distance (help traffic congestion, bad weather) customer-supplier cooperation – sole sourcing long term commitment, sole sourcing does have issues of supply disruption. Reducing machine breakdown - total productive maintenance - preventive maintenance, inspection schedule, planned shutdown for maintenance operator, operator training and job design helps in operating of any equipment in a better way and reduce breakdown incidents. Operators can be trained to do minor maintenance

work or which special person from the maintenance department is not required. Maintaining excess capacity utilization near its name plate capacity, wear and failure rates can increase rapidly with higher speed and operation and increase time of use. Operating at 80 to 90% of the name plate level, makes failures less frequent and can give higher effective capacity in the long run.

Cleanliness and housekeeping - one of the most prominent features in any WCM (world class manufacturing) facility, preventive maintenance, everything in its place and a place for everyone. Structured set ups - avoids accidental omission of any set up task. Work aids and calibration of tools and machines help in set up more accurately.

Quality - Total Quality Management

- a) product should be designed so that it is easy to make,
- b) materials from the supplier should not be defective.
- c) production should not introduce defectives,
- d) if defective items are produced, they need to be found quickly and the cause of the problem eliminated. The system should be such that production systems can be stopped to find and solve the causes of quality problems whenever they occur rather than continuing defective items and trying to rework them or work around them.

As Lean manufacturing operates with low safety stock levels, safety stocks cannot be used to make up for or buffer the company from production strategies due to defects.

Continuous improvement Kaizen, in Japanese employee suggestion system, quality circles autonomous work teams.

Employee Morale - most people like to do their job to contribute to the organization, want job security and some control over the work environment. People should be encouraged to suggest improvements in the production systems and working conditions and allowed to participate in the decision making with never possible. Lean aims to maximize the value which streams towards your customer. It considers only those processes which add value to products or services as useful. 7 kinds of waste are reduced as much as possible. Over-production, inventory, manufacturing error, manufacturing disruptions, waiting time, transport, and movements like searching for material. One way is to let the material flow through your company in as steady a stream as possible,

(flow manufacturing) with the minimum of intermediate stock and at a rate dictated by customer demand. A process flow diagram conveys which processes add value and which do not to the product, or services provided by a company.

Value Engineering or value analysis:

It is a design methodology developed in the late 1940's, which focuses on the function of the product, rather than on its structure or form.

Important aspects are:

- (i) use of multi-disciplinary teams,
- (ii) systematic procedure for evaluating product functionality and value,
- (iii) focus on product simplification. It begins by analysing the product, then each sub-assembly and finally each component if necessary.

Use of a series of questions assist the process.

The first step is to specify the function of the product and to identify those product characteristics and dimensions that are of value to the customer.

- what is the product component?
- what is its function?
- what does it do?
- why do we need to do it?
- what characteristics are of value to the customer?
- how customers will use it

Subset - how does it increase the product value.

- how this component contributes to functionality
- what does it cost?

Redesigning

- can this be done another way
- is the item over engineered?
- does it do more than necessary?
- can the item's function be combined with that of another component, how does it cost to do this?
- can a different material be used?
- can this product be made easier to assemble?

The Taguchi method: used in robust design.

Principles –

(i) when the value of a product attribute, such as shape or length, deviates from the target value, cost to its society (both consumer and producer) in terms of lower quality increases more than linearly - increases at an increasing rate (exponential).

(ii) optional design features of the product and the production process, together determine the amount of variation in the product attributes

(iii) using experimentation those products and process characteristics that affect product attributes, can be determined and by manipulating these characteristics, products can be designed to reduce the attribute variations that result from normal production variations.

Limitation is, if there are several product and process design features, each with several possible alternative levels or settings, the number of experiments is prohibitively large. This led to the development of statistical experimental design techniques to extract a maximum of information about several design features with the minimum experimentation.

Business Process Re-engineering (BPR) –

It is the radical design of business processes to achieve dramatic improvements in productivity cycle times quality and employee and customer satisfaction.

Typically, it involves the analysis of company workflows, finding processes that are sub-par or inefficient and figuring out ways to get rid of them or change them. It is the procedure of thinking and reconstructing a company's business activities to boost efficiency, reduce costs, improve product and service quality. The focus is on creating a custom centric approach to business processes with the aim of delivering products or services that meet the needs of customers in the most efficient and effective way possible.

Techniques such as process mining (the analysis of information system events logs) can help discover, monitor, and improve processes. Then it can be decided how to do the work or if it should be done at all.

Rethinking the roles of third parties or outsourcing is also a crucial component of business process reengineering – BPR.

- (a) identify and communicating the need for change - in smaller companies it is easier to look at higher user dropout and start, for a corporation it is more difficult - there will be some to be happy with things as they are, from both management and employee side, first may think of a sunk investment whereas second may be afraid of job losses. So, if the company is not doing as well as it could be, then it is not so hard to convince.
- (b) put together a team of expert: a senior manager who can call the shots + operational manager who is aware of all the process part + Re-engineering experts - right engineers from different fields, from IT to manufacturing,

- (c) find the efficient process and define Key Performance Indicators (KPI), need to define the right KPI's you don't want to adopt to a new process and then realize that you don't didn't keep some expenses in mind, idea is to optimize for manufacturing, KPI's can be - cycle time, changeover time, defect rate, inventory turnover, planned versus emergency maintenance. After defining KPI's, you will need to go after the individual process. For this, the easiest way is to do business process mapping – i) use process flowcharts or (ii) use software available for business process management,
- (d) Re-engineer the process and compare KPI's. After the analysis and planning, one can start implementing the solutions on a small scale, monitor and if the new solution works better than scaling further can be conducted, if not then go back to drawing board and start chalking up new potential solutions.

Briefly BPR is dramatic change initiative that has the following:

- refocusing company values on customer needs and eliminating low value work
- simplifying and standardizing overly complex work and automatically repetitive work,
- enabling systems with modern systems and data
- locating work in the most efficient and effective environment
- recognizing a business into cross functional teams with end-to-end responsibility for a process
- rethinking basic organizational and people issues
- determining appropriate roles for third parties or outsourcers, focusing on where they truly add value.

Example – 1980's depression

Ford Motor company started finding out inefficient process - Accounts payable division had 500 people as against their partner Mazda's who had 5. Mazda was a smaller company and Ford estimated the department was still 5 times bigger than it should have been, this led to BPR, and they could reduce it to only 100 people by reorganizing the way process was conducted. The purchasing department would write a P.O. and send a copy to Accounts Payable, material control on receiving the goods would send a copy to Accounts Payable and the vendor would also send a copy. Once all 3 copies are matched, clerk at Accounts Payable would issue the payment -> changed to purchasing issues an order and inputs it into an online database, material control receives the goods and cross references with the database to match it

with the order, then after it matches, material control accepts the order.

So, accounts payable clerk's matching the order was eliminated and could issue payment much earlier. Cycle time was reduced by 70% with faster delivery of parts to production lines, the headcount was reduced, supplier satisfaction increased from 35 to 85% and supplier's defects reduced by 90%.

Example - Airbnb - product development process

Company was struggling to find their internal identity in a design centric Silicon Valley and to create a sustainable, quick to deliver, product development process. Their product development process had three main functions – designer, engineers and researchers who worked in silos, only jumping into the process at defined times. These defined times were not sending the end goal of delivering a great product on time.

Designers had to wait on engineers to write a code before a mockup could be visualized on the screen. Engineers had to wait on researchers to validate product ideas, only to find at the very end the project assumptions were off base. It was a process failure, it needed fundamental redesigning. Head designing at Airbnb with 300 other people on product team, spent 9 months doing just that. The solution was, to create one digital environment where designer and engineer work seamlessly together. Rather than working on separate systems, a single digital environment enabled files to show updates in real time and reflect real data.

Airbnb product development process was to design product teams around outcomes do not feature. This led to teams thinking more about the customer's needs.

Organizing around outcomes, not tasks, holds true when applied to common business processes such as order-to-cash, procure-to-pay, but also to product development. At Airbnb, individual tasks (features) became the organizational priority, the larger outcome is mistakenly shifted for the immediate need of the task. Researchers who would convey at the end of a project and overturn everything, designer and developer had built, they are now participating on every stage of product development to ensure voice of guests and hosts are incorporated in.

World class manufacturing

Markets and business operations are becoming global, four forms of companies comprising WCM:

- A) Traditional - in the private sector having its own value system.
- B) Multinational - business presence in several countries
- C) Global - operations viewing the world from different angles, as in one market.
- D) Transnational - they combine benefits of global scale efficiency and of local responsiveness, examples are companies manufacturing automobiles, home appliances, mobiles etc.

Approach of all is process and results driven, guided by concepts, principles, policies, strategies, and team spirit created by corporate management system to suite the purpose, which aims at highest level of profitability besides other features.

It usually focuses on:

- (i) identification of opportunities
- (ii) creation of effective organization
- (iii) improvement/rationalization of operation
- (iv) outlining strategies and
- (v) elimination of waste.

WCM is guided by 3 principles:

1. implementing lean manufacturing and just in time principles to reduce production waste and cost. Both these go hand in hand and make sure that the items produced will meet customer demand rather than be in excess or ahead of need. Remove any manufacturing process that does not bring value to the customer. Examples: surplus inventory, lack of product demand, correcting expensive defects and unneeded transportation.
2. applying total quality management (TQM)- to decrease the number of defects and improve customer experience. TQM is based on the idea that every worker from management to frontline staff is dedicated to continuously improving the production process and creating high quality products.
3. executing total productive maintenance (TPM) - to help operation failures that can lead to defective machinery and equipment. TPM is made up of 8 pillars that focus on preventive strategies to improve equipment reliability.

Most recognized pillars of WCM are:

- a) safety - providing a safe working environment for all employees is a key pillar of WCM, established and enforce safety policies and procedures as well as provide training and resources to ensure employee safety.
- b) cost deployment - it is a systematic approach to identify and reduce cost throughout the production process

- c) autonomous maintenance - empower employees to take the ownership of the maintenance of equipment and machinery, reduce overtime and reduce cost of maintenance.
- d) professional maintenance - it ensures that all equipment is operating at optimal levels to minimize downtime and reduce costs.
- e) quality control - establish quality standards, implementing quality control measures and continuously improving quality through feedback and data analysis.
- f) logistics and material flow - to achieve high levels of productivity and minimize waste in the production process.
- g) early equipment management - to optimize the performance of equipment from the design phase to reduce downtime and improve efficiency.
- h) people development - this involves training and development opportunities, promoting a continuous improvement and empowering employees to identify and solve problems,
- i) environment - WCM recognizes the importance of environmental sustainability and implementing environmentally responsible practices.

The following can help you get started on becoming a WCM:

- develop products as solutions rather than creating ones that does not serve the needs of the customer
- exceed customer satisfaction by handling every client inquiry from the prospect stage to post-sales.
- emphasizes employee training to increase product knowledge procedures and productivity
- carry out supply chain management tactics that analyze the actual cost of a purchased product from a supplier to lessen unneeded expenses or errors

- participate in the product process (design, planning and development) protocols are followed and maintained.

Important Matrices to track are:

- overall equipment effectiveness to evaluate how efficiently operations of followed,
- overall labour effectiveness to measure the performance and quality of your work force and its productivity
- scrap rate to assess production quality and output,
- first time right to ensure workers are producing goods correctly to avoid time and money loss
- on-time orders to see how many products were delivered to customer in a timely manner,
- customer complaints to study product fulfillment.

KANBAN: Japanese word for card or ticket, KAN means visual, and BAN means card, so it refers to visual cards. Lean manufacturing uses visual cards as a signaling system that triggers an action to supply the process with its needs either from an external supplier or from warehouse. Originally invented at Toyota. KANBAN cards are often cards with data printed on them. It can vary in size, shape and content and sometimes replaced by other devices such as Golf balls, metal plates, now it is emails, sensors, electronic dashboard etc.

Example - in Hypermarket a KANBAN system is to be put in linking the display merchandise on the shelves to the warehouse and to your supplier.

The person responsible, say for a pasta, the shelf will place a Red flag on the top of the shelf when the staff needs to be replenished. The person in the warehouse will recognize this flag and will send some packs of Pasta to be displayed. The

warehouse person will need to update his warehouse data system with the withdrawal amount. The system will compare the remaining amount of Pasta to the critical ordering point, the point at which the order is to be placed with the supplier. If the level is below critical level, an email is sent to the supplier for the order automatically.

The information on the card can be:

- a part name, a part number, external supplier, or internal supplying process
- pack-out quantity, storage address, consuming process address. A barcode can be printed on the card for tracking or automatic voicing.

In a production system KANBAN has 2 functions - they visualize processes to make products and their material handlers to move products. This form is called Production KANBAN and later use is Withdrawal KANBAN.

Effective KANBAN system - Toyota had developed 6 rules for the system:

- (i) customer (downstream) processes withdraw items in the precise amounts as specified by the KANBAN,
- (ii) supplier (upstream) produces items in the process amount as specified.
- (iii) no items are made or moved without a KANBAN,
- (iv) a KANBAN should accompany each item, every time.
- (v) defects and incorrect amounts are never sent to the next downstream process.
- (vi) the number of KANBANS is reduced carefully to low inventories and to reveal problems, - visualize (the work, the workflow, and the business risks)
simplest form features: - to do,
 - in progress
 - completed
 - Limit work-in-progress
 - manage flow
 - make policies explicit
 - implement feedback loops
 - improve collaboratively, evolve experimentally.

4 principles:

- 1) start with what you are doing now.
- 2) agree to pursue incremental,
- 3) initially respect current roles, responsibilities, and job-titles

4) encourage acts of leadership at all levels

JIT - A form of production scheduling and inventory management whereby products are produced only to meet actual demand, and material for each stage of production are received or produced "just in time" for use in the next stage of production or for delivery to a customer.

2 major differences from classical production systems are:

- (i) scheduling of production relative to demand and the
- (ii) size of production batches

Classical systems, production is based on speculative demand. Marketing department forecast and production systems aligned to that, and materials ordered accordingly and first step in the production is scheduled, then subsequent steps but the execution of next step is based on completion of earlier stage of production.

In JIT, short term production is not based on demand but based on replenishing stocks depleted by actual sales and governed by inventory of a product when it reaches a certain level. So, the last stage in the production stage starts producing more of the product and inputs required for that, their production also starts. The process goes all the way to the 1st production stage and then requisition for materials signaling the suppliers for raw material.

Classical production systems produce products using longer lot sizes, longer production runs than JIT systems.

Classical system lot is based on the economic order quantity. So larger fixed costs, time of production setups, production changeovers make large lot sizes more economical by spreading those costs over a larger number of units, this leads to larger inventory and long periods of time between replenishment of stock for each product. JIT systems reduce cost and increase the system's responsiveness to demand changes by using smaller lot sizes, so lower inventory level and quicker replacement in stock of a product.

JIT systems do not operate with minimal inventories. They operate best when they are designed to operate most of the time at 80 to 90% capacity. This allows production to accelerate when demand surges. Companies operating at near capacity are forced to carry larger inventory to protect against surges in demand or other production problems, otherwise they run out of stock. When at 80 to 90% capacity, one can stop production for a while to address a problem and correct it. Examples - effective in manufacturing systems like Toyota, but even Retail Industry uses it - examples K-Mart, Walmart, Home Depot, McDonalds.

Waste Identification and Elimination:

JIT focuses on identifying all forms of waste and eliminating them. Any activity or material that costs money but does not add value to the product is eliminated.

This idea leads to inventory reduction methods such as JIT production.

Inventories do not add value to the product, but they cost money, so any decision in inventories that does not affect on-time delivery, is beneficial. Similarly quality defects are a form of waste and must be eliminated. This also is a motivator for the quality policy of doing or making things right the first time.

JIT is a management strategy that reduces waste and increases efficiency by receiving inventory only as they are needed for production, not ahead of time.

It significantly reduces 8 wastes in Lean Manufacturing:

- (i) transport
- (ii) inventory
- (iii) motion
- (iv) waiting
- (v) over production
- (vi) over processing
- (vii) defects and
- (viii) unutilized talents.

- (i) Transportation - transferring resources from one place to another. No value is added to the product as it does not directly support production. It causes several other wastes like motions and can add to waste.
 - improper facility layout or an inadequate plant design
 - poor production planning and poor scheduling
 - poor workplace organization
 - moving equipment between distant facilities

- using a faraway supplier than a nearby one

Elimination - creating an efficient workflow.

- enforcing work in progress limits

- value stream mapping

(ii) Inventory

- machinery seldom used or never used

- storage of parts that have expired or cannot be used any longer

- ordering extra parts

- over stocking

Elimination: - improve forecasting methods,

- purchasing raw material in a required quantity only when needed

- creating a queue system

- reducing buffers between production steps,

(iii) Motion - unnecessary movement of goods and employees

- Poorly located supplies and equipment

- processes taking lengthier steps than needed.

- gathering more tools than necessary

- excess activities cause the worker to search for missing tools and instructions.

Elimination - designing workstation such that it minimizes the movement.

- ergonomically placing the equipment

(iv) Waiting – interruptions in a process

- unplanned downtime idle machinery

- man-made delays

- waiting time while the machine finishes the task.

Elimination - improve communication.

(v) Overproduction - producing more than needed, faster than it is required,

- unstable scheduling

- inaccurate forecasting and demand information

- overstaffed warehouse and production facilities

- warehouse piling up the products that are not sold.

Elimination - scheduling and forecasting

- enforcing work in progress limits

- following JIT philosophy

- reduce setup times.

(vi) Overprocessing - any effect in production or communication that does not add value to the product, or the service is an overprocessing waste, also it is doing more work or more steps than required by the customer.

- unclear specifications or quality standards

- installing a part or a feature without estimating the demand

- slow decision making and approval processes.
- process bottlenecks.
- endless refinement

Elimination – understanding work requirements from a customer's viewpoint.

- use of modern tools, technologies, and processes
- use of value stream mapping

(vii) Defects - not following work instructions.

- failing to follow quality inspection process.
- product shipped to the wrong customer.

Elimination - find the route to cause of the problem using various techniques.

- design a process that can help in detecting defects during production.

(viii) Unutilized talent/skills – inoperative or misused machinery

- undertrained or mis-trained labor.
- weak or non-existent career development program
- lack of communication between leadership and staff

Elimination - using incentive for employees, coaching, mentoring.

- development streams
- technologies, management styles training methods to engage employees.

POKA YOKE:

Poka Yoke system - Japanese term that means to mistake proof a process. This is to help workers perform their job faster, work better with fewer or no mistakes, so the term devised by Japanese like old term of foolproof method to mistake proof of a process. Most common mistake is the omission of a component in an assembly or a container. Examples: Yamada Electric made a switch that required 2 springs to be inserted. The task a worker would perform was withdraw 1st spring from a large box, insert it into a switch, perform other assembly task, withdraw the 2nd spring from the box and insert it and then finish the assembly. The springs were small and once inserted, could not be seen, sometime a worker would forget whether both the springs were inserted, so will open, check, and put back again, other times worker did not notice that one spring is missing, and the switch went out defective. To overcome this issue, a change was made which increased the number of tasks and increase assembly times slightly – additional task was added to withdraw 2 springs from the box and put them on a small plate and then insert them into the switch when needed. This ensured both the springs would go in every time the plates showed 1 spring if it were not inserted from the set of two springs placed on a plate. This saved considerable time as the need to be reassemble was avoided, number of defective switches due to non-insertion of spring dropped to zero. Example - Weighing the boxes to make sure all the pieces that are supposed to be in them are enclosed.

Example - placing sensors on components so that an alarm sounds if the sensor is not in the correct position when assembled and using sensor that will stop production line if a worker hand does not withdraw a component from each box of parts.

Example – bolts intended to go in one hole can be designed not to fit in other holes by using different shapes or by putting special notching on fasteners or in holes.

Example - Nissan changed bolts on its engine mounts, gas tanks and radios so that they were tightened from above rather than below thereby making them more accessible and using rather than fighting against gravity during assembly.

EHSS management in WCM:

Environmental, Health, Safety, Security.

Protecting people and the environment is every top company's top priority, in which a comprehensive EHSS management system needs to be implemented.

Health - provide a sustainable workplace and work practices, delivering a world class performance and care.

Environment - prevent pollution and reduce waste.

Safety - provide a threat free work environment for employees.

Security - potential threats to operations, reputations, data, and assets. Many times, this may fall in or be a part of the operation management system framework. There are two basic interdependent components:

- (i) fundamentals focus attention on management principles that are most important for an effective operational management system.
- (ii) Elements - establish a structure to organize the various components of the operational management system.

Some key statistics that can be tracked:

- a) exposure hours,
- b) environment incidents, motor vehicle incidents,
- c) fire incidents,
- d) first aid provided, medical treatment,
- e) near miss recordable incident,
- f) restricted work, state warning.

Depending on the nature of a company this can be even basic lifesaving rules are provided to the employees.

Job descriptions outline agreed statement of the skills and knowledge required for effective performances in a particular job or job function and describe outcome. The primary purpose of measuring EHSS performance is to judge the implementation and effectiveness of the process established for controlling risks. This includes - gathering information on how the system operates in practice,
- identifying areas where corrective action is necessary and
- providing a basis in continual improvement.

Top management should review the EHSS system at regular intervals, to ensure its continuing suitability, adequacy, and effectiveness.

The review should include the following:

- results of EHSS management system audits,
- result from any external audits,
- communication from the interested parties,
- status of corrective and preventive actions,
- follow up action from previous reviews,
- recommendation for improvement based on changing circumstances.

Example of a company:

Goals - (1) achieve zero incident - target zero.

(2) maintains personal safety for all employees, contractors, and processes safely at all facilities.

To achieve above:

- strive for a generative EHSS culture through accountable leadership and involvement of all employees and contractors,
- set measurable objectives, KPI's, targets and action plan,
- fulfill compliance obligations, laws, regulations, internal requirements, national and international standards,
- reduce environmental footprints in alignment with example low carbon pathway, circulatory concept, positive biodiversity of ecosystem,
- empower everyone to exercise STOP WORK authority,
- commit to continuous EHSS in performance improvement.

ISO: 14001 – Environmental Management System

ISO: 45001 – Occupational Health and Safety Management.

Some principles:

- line management responsibility for safety,
- clear roles and responsibilities,
- competence commensurate with responsibilities,
- balanced priorities,
- identification of safety standards and requirements,
- hazard controls tailored to work being performed,
- operations authorization

Functions:

- define the scope of work,
- analyses the hazards,
- develop and implement hazard controls,
- perform work within controls,
- provide feedback and continuous improvement.

HR Dimensions in WCM:

With global competition, companies must now be able to effectively compete with multinational enterprises to recruit talent and maintain profitability. Given the much-reduced labor costs paid by companies in less developed nations, HR leadership must do what they can to increase throughput. So, HR teams must create the kind of work environment that can maximize employee productivity and improve employee engagement.

HR planning in WCM is expected to handle and take responsibility in the following:

- ensuring compliance and adhering to safety standards,
- harmonizing corporate culture and work conditions with company requirements,

- understanding all department's operations and staff duties,
- facilitating effective communications between floor personnel and senior management.

Strategies that are followed to build world class HR function:

1. Professionalizing each role – focus on competence, goals, technology enablement, data.
2. Clear service delivery model - creating service centres, solution centres, employee self-service and automation,
3. Design and innovation - creating designs skills, product management skills and iterative developments in HR.
4. Professional development - implementing job rotation, an HR academy, external voices, and business acumen,
5. Technology focus - creating HR Tech competence, architecture, market savvy, and close relationship with IT,
6. Employee experience - understanding roles, journeys, transitions, wellbeing, and all aspects of employee productivity,
7. having a world class competence in Talent acquisition, development and mobility, rewards, talent analysis,
8. Operating as a team - creating an integrated HR strategy, making sure people know each other, help each other, and rotate within and in and out of HR.

Maintenance Practices:

Work stoppages and product defects are often the result of poor equipment maintenance. So, a major aspect in WCM, is a system for improving maintenance and reducing machine breakdowns.

Total Productive Maintenance or TPM, is the maintenance strategy often followed.

One of the primary causes of inadequate preventive maintenance is that it is assigned to a specialized maintenance crew, these personnel are expensive, so most companies tend to keep their number smaller, and give repair of breakdowns, the highest priority. So, any time left for preventive maintenance is utilized for major work and as overhauls that is considered more important, leaving routine preventive maintenance no time and this leads to frequent breakdowns. A solution to this is to train machine operators to perform some of their own maintenance, especially preventive maintenance, and it can be done when the machine is running or while it is idle and the maintenance costless.

Operating at 80-90 % of the name plate level, makes failures less frequent than operating on near capacity and though in short period, one may find counterproductive but in the long run it produces more output, simply due to less breakdowns, stoppage of work due to defects. Another factor that is seen in WCM is seen, cleanliness on the shop floor. A major cause of machine failures and poor quality is the buildup of dust, oil, and chemicals in machines or on materials. Cleaning machines should be mutual part of the operating personnel and can be done easily in idle times. Tools kept in a designated place, materials arranged neatly, aisles kept clear of clutter, keep a work atmosphere pleasant and attractive one; improves visibility and sight lines and reduces accidents that can injure employees and damage machines. Having a structured set up guarantees no setup task is omitted by accident. In addition, work aids, calibration of tools and machines make it more likely that setup will be done accurately. Human errors can be minimized by employing the job design principles and work aids, like guide bars, stop bars and various POKA YOKE devices (mistake proofing).

Several other principle and actions can enhance equipment maintenance:

- 1) machines should be kept as simple as possible while still accomplishing their required functions, standardizing replacement parts, and reducing variety of machines for maintenance personnel ,
- 2) collection data on the frequency and causes of machine failure, to allocate maintenance resources efficiently,
- 3) when some parts of machine wear out, many other components are also wearing out as well. It is better to rebuild the machine by replacing all worn out parts at once, instead of replacing them one by one,
- 4) key components should be produced by careful planning, so that all parts are on hand when needed.

WCM in reference to Indian industry and Indian scenario:

India in comparison with China was seen earlier as an economy where risk is higher and the business environment more problematic than in rival Asian investment locations.

Currently the scenario has changed, the legal framework that protects investment is one of the best in Asia. The economy offers an abundance of technical and managerial talent, often with international experience. Geopolitical risk is diminishing consistently, especially with respect to China. India has a demographic advantage that should see its working age population continue to grow well into the century, increasing wealth and reducing cost. The companies range from Bharat Forge, Bajaj, and Tatas, in auto sector, To L&T and Godrej in specialist engineering. Ballarpur Industries on paper, quite a good number in pharmaceuticals in chemicals as well Moser Baer. And they are showing that India is beginning to shrug off its reputation of appalling quality and reliability, and that it can be compete internationally.

Indian industry is also emerging as an outsourcing design and production based for manufacturing as it has been for software, many of the auto companies and now mobile companies are sourcing components and assembling products.

Tata Motors does its own design, but 74% of its work is for foreign clients like Chrysler, General Motors, Boeing, Airbus etc. Cummins is making India its manufacturing hub for newly developed line of generator sets, Samsung making a plant in Chennai as its global hub, Ford for engine manufacturing, Suzuki and Hyundai for manufacturing and export hub for small cars, all 5 top telecom manufacturers have set up manufacturing facilities in India.

Several multinational operating in skill-intensive industries requiring advanced technical expertise have set their shops in India. ABB, Honeywell and Siemens in electrical and electronic products, Cummins, Daimler, Chrysler, Toyota Motor in auto components and engineering, Degussa as well as Rohm and Haas in specialty chemicals have all set up the manufacturing house in India. Indian companies are also becoming renowned for their adherences to global quality standards.

The Government has also taken initiatives to accelerate growth and improve competitiveness of Indian industry in general, by implementing technology upgradation schemes for small scale industries, textiles, food processing among others, infrastructure upgradation on cluster basis, easier access to inputs, reduction in duty rates, encouragement to foreign technology, collaborations, liberalization of FDI in manufacturing activities, special economic act.

10 key sectors that comprise bulk of Indian manufacturing are Engineering, Electronics, Automotives, Textiles, Chemicals, Leather, Metals, Machines Tools, Food Processing, Gems and Jewelry.

Largest 3-wheeler market, 2nd largest in 2-wheeler, 2nd largest in jewelry market, largest diamond cutting and polishing center, 2nd largest in cotton trade. Chemical Industry ranks 12th by volume, largest producer of milk, tea and pulses with world's largest livestock to support food processing sector, 5th in bauxite reserves.

Top Companies: Maruti Suzuki, Unilever, Mahindra, Apollo Tyres, Bajaj Auto, Hero Motocorp, Tata Motors, Ashok Leyland, Dabur India, Reliance Industries, Sun Pharma, Hindalco, BPCL, HPCL, Indian Oil Corporation.

Examples: Vedanta has taken a digitalization across the value chain and started incorporating technology to mitigate risks and minimize environmental impact while constructing tailing dams, Hindalco has developed an effective tool - internal carbon pricing, to enable more carbon efficient projects to get approved through the investment decision process. This mechanism helps to monitor the cost of carbon, prepare the Hindalco for upcoming regulations, emission trading schemes, carbon taxes etc.

The benefits realized by going WCM way:

- 1) increase in brand value - being proactive towards sustainability, helps organization to be ready for the related regulations, reduce compliance-

related risks, minimize climate and supply chain risks, builds and improves brand image, customer loyalty, investor relations.

Example – Hindalco was in 2021, the world's most sustainable aluminium company conferred by Dow Jones Sustainability Indices (DJSI), for 2nd consecutive year and it is the only aluminium company which is the part of DJSI world Index.

- 2) Companies now are finding ways and means to utilize the waste generated from factories by partnering with recyclers. The utilizing waste as a raw material for other industries, thereby building ecosystems which will help to achieve circular economy - example of cement industry<->iron waste,
- 3) Increase in bottom line - reduced business costs, more innovative strategies, an improved reputation, new business models, new customers who value sustainability leading to more earnings and direct impact on the bottom line of the organization.

Through JIT concept has been implemented in manufacturing organization, these are completely opposite to traditional ways of thinking. So Indian companies must be willing to modify their procedures and operations. Training of employees, procedures of dealing with suppliers, simplification, automation, and engineering of operational process need to be examined prior to implementation of JIT. Here in India, social, cultural, and political matters have a significant impact on JIT practices in different parts, suppliers of raw materials (imported and domestic) are subjected under Government control through supply agencies, giving rise to high uncertainty. Government's central pricing of key resources and taxation rates, all create obstruction in the way of implementation of WCM practices. The Indian labour, unskilled or the low skilled version is highly uneducated, lacking in motivation and more concerned with monetary benefits and job security than career progress and development of the potential. Labor unions and their reluctances are also unfavorable in WCM. Therefore, specific cultural changes are required for getting converted or attaining a WCM standard, quite often it is seen Japanese Training models are not highly successful in India, so need to adapt or develop modules as per Indian workforce after studying the behaviour patterns, personal traits, attitudes, social values etc.

Industry 4.0 is a holistic automation, business information and manufacturing execution architecture to improve industry with the integration of all aspects of production and commerce across company boundaries for greater efficiency, IoT - internet of things is more important aspect of it. IoT market in India is expected to have close to 20% + share of the Global market. Andhra Pradesh has taken the initiative to capitalize on IoT potential in the country. The Indian government has created green energy corridor to bring in more renewable energies to make

smart grids that will support the variable input and create stage in in AP, Gujarat, Maharashtra, Tamilnadu, Himachal Pradesh, Rajasthan.

India's first smart factory is in Bengaluru, making progress at Indian Institute of Science Centre for Product Design and Manufacturing, with an investment from Boeing. A smart factory armed with data exchange in manufacturing and the Internet of Things (IoT), is the future and experts are calling it revolution industry 4.0.

Bajaj Auto initiated in 2010, the process of automation and uses 100-120 Cobots (collaborative robots) in its production facilities. Maruti Suzuki has 7 process shops and 5 assembly lines with around 1700 robots. Ford at Sanand, has 437 robots, Hyundai at Sriperumbudur reduced labor units by utilizing 400 robots, Tata Motors has over 100 robots in Sanand plant.

While industry 3.0 was simply about automation of isolated machines, Industry 4.0 concentrates on the end to end digitization of all physical assets and their integration into digital ecosystems with value chain partners. This is leading to crowding out effect in addition to increasing competition from emerging manufacturing bases like Vietnam, Türkiye, and Taiwan. At the same time, it gives India, a second destination for other countries.